

Layer-wise Analysis of Representation Fine-Tuning under resource constraints: Which layers matter most?

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Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

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I. INTRODUCTION

Since the introduction of transformers [1], the landscape has changed to building large-scale language models. Today, Large Language Models (LLMs) are comprised of billions of parameters [2]. With this enormous size comes the challenging task of fine-tuning these models for specific tasks. While full finetuning still yields more accurate results, the cost of this is enormous. To address this challenge, the field of parameter-efficient finetuning (PEFT) has gathered considerable attention. These methods enable researchers to experiment on state-of-the-art models with limited computational resources.

Different techniques have been introduced to fine-tune LLMs. Adapter layers insert a small module in a frozen transformer and train it [4]. Prefix tuning modifies the inputs to attentions instead of modifying weights or activations [5]. Hu et al. [3] introduced a Low-Rank Adaptation (LoRA) of Large Language Models, a way to fine-tune a model without fine-tuning all of its parameters. All these methods either change the inputs or modify the weights without requiring full finetuning.

One more highly efficient technique of PEFT is Representation Finetuning (ReFT) for Language Models. In ReFT, the representations are directly modified instead of weights of the models [6]. Hence, ReFT acts as a drop-in intervention in representations and steer the output of the model towards a specific direction. Furthermore, ReFT trains over less parameters compared to a standard LoRA setup and achieves fairly competent performance over various tasks.

Unlike weight-based adapters which modify model's knowledge base, ReFT operates on the hidden states. ReFT helps the model learn geometric transformations which in result nudge the direction of the model's output without altering its memory. This distinction is crucial because it signals the success of these representations depends more on the semantic richness of

the representations at a given depth, which can vary depending on the model architecture and the task of finetuning.

However, ReFT finetuning has a large hyperparameter search space and it is sensitive to layer selection [6]. Also, if ReFT is applied to multiple layers, the intervention matrices must be kept active during inference, introducing some inference delay. Furthermore, ReFT has not been thoroughly studied for Small Language Models (SLMs) under resource constraints on a limited dataset. The original paper also showcases how ReFT is susceptible to nature of tasks as well, further expanding the hyperparameter search space. This requires investigation on how the search space can be narrowed down based on the model architecture and the dataset constraints in place.

With this backdrop, it can be highlighted that the efficacy of ReFT is regulated by layer-wise specialization and architectural constraints. Existing work has primarily focused on 7B+ models. The behavior of ReFT in the 'Low-Resource' areas is under-explored. The research will focus on the following questions:

- What does the trend of loss looks like for the layers? Do some layers tend to overfit in ReFT while other do not? Does the training comply with existing literature on layer fine-tuning?
- Does the accuracy change drastically between the layers to specific tasks?
- Are interventions productive for small language models?
- Are the accuracy and representation trends between the models similar across all layers?

II. LITERATURE REVIEW

A. Representation Finetuning

Representation Finetuning (ReFT) is a parameter-efficient finetuning technique for LLMs [6]. It trains interventions over the hidden states to steer model behavior into the direction of the finetuning task at inference time. Unlike other methods that target weights, ReFT modifies the hidden representations. The main inspiration of this methodology is the study of causal interventions to direct the behavior of the model [7].

There are two types of interventions introduced originally: LoReFT and DiReFT. LoReFT learns intervention in linear subspace spanned by a low-rank projection matrix, while DiReFT simply shifts the representation by a fixed vector. DiReFT is more straightforward and efficient but is not as accurate as LoReFT is. LoReFT transformation looks like this:

$$\Phi(h) = h + R^T(Wh + b - Rh)$$

Where h is the original hidden representation, R is the low-rank projection matrix, W is the learned weight matrix and b is the learned bias vector.

B. Linearly Decodable Features

Research has shown that many features encoded in the hidden representations are linearly decodable, simulated with linear probing experiments [8]. This is the area of interpretation that ReFT targets. When features are rather linearly decodable, a learned linear projection over hidden representations can be quite effective. However, this is not always the case, as other literature argues not all the features are linearly decodable [9]. This makes ReFT potentially less effective where the relevant subspace cannot be manipulated through linear interventions, varying its efficiency from task to task.

C. Layer-wise Analysis

Much work has shown how layers

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A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
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$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

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Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

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- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
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- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

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TABLE I
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	<i>Table column subhead</i>	<i>Subhead</i>	<i>Subhead</i>
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ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

IV. REFERENCES

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